

Document Control

- System Name: Application

Executive Summary

- This BRD details business requirements for ingesting customer and order CSV data into Delta tables, applying cleansing rules, computing a TotalAmount, creating an SCD Type 2 ordersummary, and aggregating daily customer spend, all orchestrated via Delta Live Tables.
- System Name: Application

Business Objectives

- Enable reliable ingestion of customer and order CSV data into Delta tables.
- Enforce data cleansing by removing "Null"/Null literals and duplicates.
- Enrich order records with TotalAmount ($\text{PricePerUnit} \times \text{Qty}$).
- Establish ordersummary table with SCD Type 2 change management for customers.
- Aggregate spend per customer per date in customeraggregatespend.
- Standardize pipeline steps using Delta Live Tables.

In-Scope

- Ingestion of customer and order CSV files from defined paths.
- Creation and population of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Application of data cleansing rules.
- Calculation of TotalAmount.
- SCD Type 2 history management in ordersummary.
- Aggregation to customeraggregatespend.
- Implementation using Delta Live Tables.

Out-of-Scope

- Integration with non-listed data sources or fields.
- Security, access controls, monitoring, and alerting.
- Performance optimization and scaling.
- Scheduling and DevOps outside Delta Live Tables steps.
- Data quality rules beyond specified cleansing.
- User interfaces, analytic dashboards, report layers.

Stakeholders

- Information not present in source JSON.

Business Requirements

BR-ING-001 — Source Data Ingestion

- Description: Load customer and order CSV files into corresponding Delta tables from specified filesystem paths.
- Business Value: Ensures controlled and auditable data onboarding pipeline.
- Priority: High

BR-SCH-002 — Schema Enforcement

- Description: Apply defined schemas to customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId) tables.
- Business Value: Enforces structure and supports data integrity.
- Priority: High

BR-CLE-003 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables.
- Business Value: Delivers quality inputs for downstream analytics.
- Priority: High

BR-COM-004 — Computed TotalAmount

- Description: Calculate TotalAmount as $\text{PricePerUnit} \times \text{Qty}$ for each order.
- Business Value: Adds essential metric for spend analysis.
- Priority: High

BR-JOI-005 — Customer-Order Join

- Description: Join customer and order data on CustId to create an ordersummary table.
- Business Value: Centralizes transactional context for analysis.
- Priority: High

BR-SCD-006 — SCD Type 2 Customer Dimension

- Description: Implement SCD Type 2 logic in ordersummary for historical customer attribute management (Active/Inactive, StartDate/EndDate, full history).
- Business Value: Supports time-travel reporting and regulatory compliance.
- Priority: High

BR-AGG-007 — Customer Aggregate Spend

- Description: Compute and store daily aggregate spend (sum of TotalAmount by Name and Date) in customeraggregatespend table.

- Business Value: Enables trend and behavior analytics.
- Priority: High

BR-DEP-008 — Delta Live Tables Implementation

- Description: Execute the entire pipeline (steps 1–10) using Delta Live Tables as orchestration and ETL engine.
- Business Value: Ensures reliability, automation, and alignment with modern data stack.
- Priority: High

Assumptions & Constraints

- Only requirements stated in the source JSON are assumed; all unspecified elements are not included.
- All pipeline steps must utilize Delta Live Tables.
- SCD Type 2 metadata fields are referenced but definitions are not explicit in JSON.
- No additional logic, partitioning, or indexing unless stated in requirements.

Success Metrics

- 100% successful ingestion of customer and order source files with schema enforcement.
- All records cleansed of "Null"/Null literals and duplicates before transformation.
- Orders enriched with TotalAmount as per expression.
- Accurate SCD Type 2 implementation in ordersummary.
- customeraggregatespend contains correct daily sums for each customer.
- End-to-end execution through Delta Live Tables with no data loss.

Risks & Mitigations

- Ambiguity around SCD Type 2 attribute field names: Flag for further clarification.
- Insufficient detail on error handling or recovery: Recommend adding explicit requirements.
- Potential for data quality issues not addressed by cleansing logic: Perform downstream data audits.
- Absence of explicit performance, scheduling, or access requirements: Identify and define before implementation.

Approval

- Section for sign-off and stakeholder acceptance (to be completed during review).

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
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Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into customer Delta table
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into order Delta table

Appendix B: Target Definitions

Layer	Table	Columns	Notes
Raw	customer	CustId, Name, EmailId, Region	Ingested and cleaned
Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Ingested, cleaned
Processed	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Type 2 loaded
Aggregate	customeraggregatespend	Name, TotalAmount, Date	Daily spend summary

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Load CSV	customerdata	customer	Schema applied, cleanse
2	Load CSV	orderdata	order	Schema applied, cleanse
3	Compute	order	order	TotalAmount = PricePerUnit × Qty
4	Join	customer, order	ordersummary	Join on CustId, schema as specified
5	SCD Type 2	ordersummary	ordersummary	Activate/expire records, maintain history
6	Aggregate	ordersummary	customeraggregatespend	sum(TotalAmount) by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-1	Computation	TotalAmount = PricePerUnit × Qty	Requirements
BR-2	Cleansing	Remove "Null"/Null literals; Dedupe	Requirements
BR-3	Join	customer to order on CustId	Requirements
BR-4	SCD Type 2	Close old, activate new, track dates	Requirements
BR-5	Aggregation	sum(TotalAmount) by Name, Date	Requirements

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer/order data from CSV	None
2	Apply schema, cleansing, computed columns	Step 1
3	Join for ordersummary with SCD Type 2	Step 2
4	Aggregate to customeraggregatespend	Step 3
5	Persist all outputs to Delta tables	Step 4

Appendix F: Entity Relationship Diagram (ERD)

The following text-based ERD describes the data model structure:

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

+CustId

TotalAmount

[ordersummary]

*OrderId

CustId

Name

EmailId

Region

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

[customeraggregatespend]

*Name

Date

TotalAmount

customer 1--* order

customer 1--* ordersummary

order 1--* ordersummary

ordersummary *--* customeraggregatespend